Adaptive Topology Routing for Game-Theoretic Multi-Agent LLM Consensus via AEIC Decomposition

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Abstract. Multi-agent large language model (LLM) systems achieve collective reasoning through output aggregation, yet existing consensus mechanisms lack formal incentive guarantees and treat communication topology as a fixed design choice. We propose a framework combining three innovations: (1) AEIC (Assumption–Evidence–Inference–Conclusion) decomposition that structures agent reasoning into auditable components with four computable quality signals; (2) a Stackelberg incentive mechanism with 16 learnable parameters optimized via bilevel differential evolution with cross-validated evaluation, provably achieving ϵ-incentive compatibility; and (3) an adaptive neural topology router that selects among five communication topologies per question using AEIC-derived features. On three benchmarks (633 questions evaluated across all 5 topologies, 5 agents), our router achieves 87.6% consensus accuracy—+4.5pp above the best fixed topology (star, 83.1% on the same evaluation set). A key finding is that AEIC quality signals measure reasoning process quality rather than answer correctness: AEIC-weighted voting is statistically no better than majority voting (significantly worse on 3 of 5 topologies, McNemar $p<0.05$). However, these signals are powerful features for meta-level topology selection, closing 46.5% of the gap to the oracle ceiling (92.9%). The framework additionally supports online Stackelberg feedback, where agents receive quality assessments between rounds; this yields up to +6.7pp improvement on knowledge-intensive debate tasks, though effects are topology- and task-dependent.

Keywords: Multi-agent consensus · Stackelberg game · Topology routing · Large language models · Incentive compatibility

1 Introduction

Multi-agent systems powered by large language models (LLMs) have demonstrated strong potential in complex reasoning tasks through collective deliberation [11,21]. By deploying multiple LLM instances to independently analyze a problem and aggregating their outputs, these systems can surpass single-agent performance on benchmarks spanning commonsense reasoning, mathematical problem-solving, and factual question answering [22].

However, two fundamental tensions underlie multi-agent LLM collaboration. First, *how should the system aggregate diverse agent outputs into a reliable consensus?* Current approaches—majority voting [22,26] and multi-round debate [8,6]—lack formal incentive guarantees. Agents are not formally motivated to exert effort in producing high-quality reasoning, creating a *moral hazard* where downstream agents may sycophantically adopt upstream conclusions [16,2].

Second, *how should agents communicate?* The communication topology—whether agents reason independently (parallel), observe predecessors (sequential), aggregate through a hub (star), exchange iteratively (debate), or operate in hierarchical groups—fundamentally affects both the quality of individual reasoning and the reliability of collective judgment. Yet existing work treats topology as a fixed architectural choice rather than an adaptive decision conditioned on problem characteristics.

We address both problems through a unified framework that combines game-theoretic consensus with adaptive topology selection. Our contributions are:

1. **AEIC Decomposition as Dual-Purpose Signal Extraction.** We decompose each agent's reasoning into four auditable dimensions—Assumption (A), Evidence (E), Inference (I), and Conclusion (C)—yielding quality signals (IC, EC, EQ, EV) that serve both as inputs to incentive mechanism design and as rich features for topology routing (Section 3).
2. **Stackelberg Incentive Mechanism.** We formulate consensus as a Stackelberg game with 16 learnable parameters optimized via bilevel differential evolution. The mechanism provably achieves ϵ-incentive compatibility with a controllable accuracy–incentive trade-off. Per-topology optimization discovers interpretable parameter profiles, including an anti-sycophancy effect where cross-agent consensus signals are suppressed (Section 4).
3. **Adaptive Topology Routing with Neural Scorer.** We introduce a neural topology router that selects from five communication topologies per question using 42-dimensional AEIC-derived features per topology. With per-dataset specialized scoring heads, the router achieves 87.6% accuracy—closing 46.5% of the gap between the best fixed topology (83.1%) and the oracle ceiling (92.9%) (Section 5).

A central finding is that AEIC quality signals measure *reasoning process quality* rather than *answer correctness*: majority voting outperforms AEIC-weighted aggregation on 4 of 5 topologies. This reveals that the signals' value lies in meta-level topology selection rather than direct vote weighting (Section 7).

2 Related Work

Multi-Agent LLM Consensus. The aggregation of multiple LLM outputs has been studied through several paradigms. Self-consistency [22] samples multiple reasoning paths and selects the most frequent answer. Multi-agent debate [8,6] extends this to multiple model instances engaging in iterative discussion. ChatEval [3] applies multi-agent debate to evaluation tasks. Pitre et al. [16] address

sycophancy through specialized agent design. Zhao et al. [26] propose cardinal voting based on social choice theory. Li et al. [13] introduce dynamic weighting. Bertalanič and Fortuna [2] quantify the cost of consensus, reporting sycophancy rates up to 85.5% in multi-round settings with Qwen2.5-7B. None of these methods provides formal game-theoretic incentive guarantees or considers adaptive topology selection.

Communication Topology in Multi-Agent Systems. The effect of communication structure on multi-agent performance has been explored in several contexts. Chen et al. [5] use a centralized evaluator (star topology). Sequential architectures appear in chain-of-agents designs [12]. Debate topologies (complete graphs) are studied by Du et al. [8]. Chen et al. [4] introduce round-table conference among diverse LLMs with confidence-weighted voting. Most recently, RUMAD [1] applies reinforcement learning to adaptively control debate rounds and agent selection, reporting token cost reductions while maintaining accuracy. CARD [15] proposes customizable routing across debate structures. However, these approaches either fix the topology or adapt only within a single topology type (e.g., varying debate rounds). We consider adaptive selection across five fundamentally different communication structures using learned quality signals, which is complementary to within-topology adaptation.

Game Theory and LLMs. Sun et al. [20] provide a taxonomy of game theory applied to LLMs. Yi et al. [25] model multi-agent debate as a Bayesian game with Nash equilibria. Rauba et al. [17] argue for principal–agent modeling of LLM systems. Liu et al. [14] use LLMs for automated mechanism design. Our work differs by using a Stackelberg (leader–follower) framework rather than simultaneous-move equilibrium, which is more natural for mechanism design where the aggregation rule is committed before agents act.

Stackelberg Games in Distributed Systems. Stackelberg incentive mechanisms are established in federated learning [9,10] and crowdsourcing [23]. Saig et al. [18] apply contract theory to incentivize quality text generation from LLMs. Xu et al. [24] apply Stackelberg games to multi-agent contract revision. To the best of our knowledge, no prior work applies Stackelberg mechanism design to multi-agent LLM consensus with adaptive topology routing.

3 AEIC Decomposition and Quality Signals

We introduce the AEIC decomposition that provides auditable reasoning quality signals for both mechanism design and topology routing.

3.1 Structured Reasoning Decomposition

Given a question x, each agent $i \in \{1, \ldots, n\}$ produces a structured response decomposed into four components:

- **Assumption (A)**: The premise or hypothesis the agent adopts.
- **Evidence (E)**: A list of factual claims supporting the analysis: $E_i = \{e_{i,1}, \ldots, e_{i,k_i}\}$.
- **Inference (I)**: The reasoning chain connecting evidence to conclusion.
- **Conclusion (C)**: The final answer with justification.

This decomposition is enforced through prompt engineering and serves two purposes: (1) it structures the reasoning process to improve output quality, and (2) it provides separable components whose quality can be independently assessed by the consensus mechanism.

3.2 Four-Dimensional Quality Signals

From each agent's AEIC output, we compute four quality signals using embedding-based similarity:

Definition 1 (Quality Signals). *Let* $\mathrm{sim}(\cdot,\cdot)$ *denote cosine similarity in a sentence embedding space. For agent* i*:*

$$\mathrm{IC}_i = \mathrm{sim}(I_i, C_i) \tag{1}$$

$$\mathrm{EC}_i = \tfrac{1}{|E_i|} \textstyle\sum_{e \in E_i} \mathrm{sim}(e, C_i) \tag{2}$$

$$\mathrm{EQ}_i = f_{\mathrm{quality}}(E_i) \tag{3}$$

$$\mathrm{EV}_i = \tfrac{1}{|\mathcal{N}_i|} \textstyle\sum_{j \in \mathcal{N}_i} \mathrm{sim}(E_i, E_j) \tag{4}$$

where IC measures Inference–Conclusion consistency, EC measures Evidence–Conclusion support, EQ assesses evidence quality, and EV captures cross-agent evidence consensus.

Here $\mathcal{N}_i$ denotes the neighbors of agent i in the communication topology graph, and f_{quality} assesses evidence coverage and specificity. IC, EC, and EQ measure *individual reasoning quality*: how coherent, well-supported, and thorough an agent's analysis is. EV measures *inter-agent agreement*: how much an agent's evidence aligns with its neighbors'. This distinction is critical for incentive design—EV can be inflated by sycophantic behavior, while IC/EC/EQ require genuine reasoning effort.

3.3 Agent Type Space

Following Saig et al. [18], we model each agent through an observable *type*:

Definition 2 (Agent Type). *The type of agent* i *on question* q *is:*

$$\tau_i = (IC_i, EC_i, EQ_i) \in [0,1]^3 \tag{5}$$

EV is deliberately excluded from the type. It measures consensus rather than reasoning quality—a sycophantic agent can achieve high EV by mimicking neighbors without genuine reasoning. The type captures what the mechanism should reward: the quality of individual reasoning effort.

3.4 AEIC as Feature Extraction

Beyond mechanism design, AEIC signals serve as feature representations for topology routing. For each question–topology pair, we construct a 42-dimensional vector from per-agent signals, summary statistics, and cross-signal interactions. The same agents produce different signal profiles under different topologies, enabling a learned router to identify optimal communication structures (Section 5).

4 Stackelberg Consensus Game

4.1 Game Formulation

We formulate multi-agent consensus as a Stackelberg game $\mathcal{G} = (\mathcal{L}, \mathcal{F}, \Theta, \mathcal{S}, U_L, U_F)$.

Players. The *leader* $\mathcal{L}$ is the mechanism designer (bilevel optimizer). The *followers* $\mathcal{F} = \{1, \ldots, n\}$ are the LLM agents.

Strategy Spaces. The leader chooses mechanism parameters $\theta \in \Theta \subset \mathbb{R}^{16}$, controlling how AEIC quality signals map to consensus weights. Each follower i chooses a strategy $s_i \in \{s^{\text{ind}}, s^{\text{syc}}\}$:

- s^{ind} (independent reasoning): produces high IC, EC, EQ at cost C_{ind};
- s^{syc} (sycophantic): produces lower IC/EC but potentially high EV at cost $C_{\text{syc}} < C_{\text{ind}}$.

Timing.

1. Leader commits to parameters θ (mechanism rules).
2. Each follower observes θ and chooses strategy s_i.
3. Agents produce AEIC outputs; quality signals τ_i are computed.
4. Mechanism computes weights $w_i(\tau_i; \theta)$ via the scoring rule.
5. Weighted voting produces the consensus decision.

Utility Functions. The follower utility is:

$$U_i(s_i; \theta) = w_i(\tau_i(s_i); \theta) \cdot R - C(s_i) \tag{6}$$

where $R > 0$ is the reward for adoption. The leader maximizes:

$$\max_{\theta \in \Theta} \text{Accuracy}(\theta) + \lambda \cdot \text{IC}(\theta) \tag{7}$$

4.2 Scoring Rule

The 16 parameters θ control two aspects:

- **Bid construction** (12 parameters): How IC, EC, EQ, EV signals combine into agent "bids" (alignment willingness α_i, revision cost, historical alignment h_i).

– **Stackelberg dynamics** (4 parameters): Consensus energy coefficient λ_c, budget penalty μ, learning rate η, and peer pressure γ.

Specifically, for agent i:

$$\alpha_i = \max\big(\theta_{\text{floor}},\ \text{IC}_i^{\theta_{\text{ic}}} \cdot \text{EQ}_i^{\theta_{\text{eq}}} \cdot (\theta_{\text{ev_base}} + \theta_{\text{ev_scale}} \cdot \text{EV}_i)\big) \tag{8}$$

$$h_i = \theta_{h_1} \cdot \text{IC}_i + \theta_{h_2} \cdot \text{EC}_i + \theta_{h_3} \cdot \text{EQ}_i + \theta_{h_4} \cdot \text{EV}_i \tag{9}$$

These bids are processed through a distributed consensus game that produces final weights $w_i(\theta)$ via iterative updates incorporating peer pressure from a graph Laplacian defined by the communication topology.

4.3 Incentive Compatibility Metric

We define incentive compatibility through inter-group weight comparison:

Definition 3 (Incentive Compatibility Metric). *For each question q, rank agents by quality score $\phi(\tau_i) = (IC_i + EC_i + EQ_i)/3$. Let H_q = top-2 agents and L_q = bottom-2 agents. Define:*

$$IC(\theta) = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{1}\left[\frac{1}{|H_q|} \sum_{i \in H_q} w_i(\theta) > \frac{1}{|L_q|} \sum_{j \in L_q} w_j(\theta)\right] \tag{10}$$

This measures the fraction of questions where higher-quality reasoning receives greater consensus weight, capturing the essence of incentive compatibility.

4.4 Bilevel Optimization

The Stackelberg game is solved via bilevel optimization:

$$\textbf{Upper (Leader):} \quad \max_{\theta \in \Theta} \ \text{Accuracy}(\theta) + \lambda \cdot \text{IC}(\theta) \tag{11}$$

$$\textbf{Lower (Followers):} \quad w_i = f(\tau_i; \theta) \quad \forall i \tag{12}$$

The lower-level problem is deterministic given θ (agents' AEIC outputs are fixed offline), reducing to single-level optimization implementable via differential evolution [19]. Per-topology θ_T^* is optimized with population size 16 (1× the number of parameters) and 30 generations, then evaluated via 5-fold stratified cross-validation to prevent train/test leakage.

4.5 Online Stackelberg Extension

In multi-round topologies (debate, hierarchical), the mechanism operates *online*: between rounds t and t+1, it computes interim AEIC signals from round-t outputs, runs the Stackelberg game to obtain current weights $w_i^{(t)}$, and injects quality feedback into agent prompts:

$$\text{context}_i^{(t+1)} = \text{context}_i^{(t+1)} \cup \{(\text{IC}_i^{(t)}, \text{EC}_i^{(t)}, w_i^{(t)})\} \tag{13}$$

This transforms the game from post-hoc reweighting to real-time incentive shaping: agents observe their current quality assessment and can adjust reasoning effort accordingly. The star topology is inherently online—leaf trust scores computed via Stackelberg are injected into the hub's synthesis context.

5 Adaptive Topology Routing

A key insight of this work is that no single communication topology dominates across all questions. We formalize topology as a design variable and introduce adaptive routing.

5.1 Communication Topologies

We define five communication topologies as weighted adjacency matrices $W \in \mathbb{R}^{n \times n}$:

- **Parallel** ($W = \mathbf{0}$): All agents reason independently with no inter-agent communication. This is the standard self-consistency setting.
- **Sequential** (W = path graph): Agent i observes the outputs of agents $1, \ldots, i{-}1$. Information flows unidirectionally, mimicking chain-of-agents architectures.
- **Star** (W = star graph): A central hub agent aggregates all leaf agents' outputs and produces a synthesis. Leaf agents reason independently.
- **Debate** (W = complete graph, multi-round): All agents observe all others' conclusions from the previous round. Repeated for 3 rounds, simulating multi-agent debate.
- **Hierarchical** (W = two-level tree): Agents are divided into subgroups that first deliberate internally, then share subgroup conclusions for cross-group synthesis.

The topology W affects both signal computation (EV and agreement signals depend on $\mathcal{N}_i$) and agent behavior (context-aware generation in sequential, star, debate, and hierarchical modes).

5.2 Per-Topology Parameter Optimization

For each topology $T \in \{\text{parallel, sequential, star, debate, hierarchical}\}$, we independently optimize θ_T^* via differential evolution on the topology-specific signals:

$$\theta_T^* = \arg\max_{\theta \in \Theta} \ \text{Accuracy}_T(\theta) + \lambda \cdot \text{IC}_T(\theta) \tag{14}$$

This yields five specialized consensus configurations. The optimized parameters reveal topology-specific structure (Figure 3): star and debate topologies learn near-zero weights for EV ($w_{\text{EV}} \approx 0.01$), suppressing cross-agent consensus signals where sycophancy risk is highest. Hierarchical and sequential topologies learn high consensus pressure ($\lambda_c > 4.5$), enforcing agreement within their chain/tree structures.

5.3 Neural Topology Router

To select the optimal topology per question, we train a neural router using AEIC-derived features.

Feature Design. For each candidate topology T, we construct a 42-dimensional feature vector comprising per-agent signals (IC, EC, EQ, EV, agreement for 5 agents), summary statistics (mean, std, range), voting features (majority ratio, yes/no ratio), and cross-signal interactions (*e.g.*, $\text{IC}_i \times (1-\text{agreement}_i)$). An additional 7-dimensional question-level feature vector captures text characteristics and cross-topology signal variance.

Router Architecture. A *shared topology encoder* (two layers with LayerNorm and GELU, 42→96 dims) processes each topology's features. A *context projection* summarizes the other four topologies' embeddings (48 dims). Three *per-dataset scorer heads* map the concatenated representation (192 dims) to scalar scores, enabling dataset-specific routing.

Training. We use a focal ranking loss with adaptive margins:

$$\mathcal{L} = \sum_{(c,w)} \left(\frac{v}{m}\right)^{\gamma} \cdot v, \quad v = \max(0,\ m - (s_c - s_w)) \tag{15}$$

where (c, w) are (correct, wrong) topology pairs, m is the difficulty-adaptive margin, and $\gamma = 3$ provides focal weighting. Training uses AdamW with cosine annealing over 300 epochs, 5-fold stratified cross-validation.

Top-1 Selection. The router selects the highest-scoring topology per question as the consensus topology. With per-dataset specialized scoring heads, this achieves 87.6% ± 0.3% accuracy across 5 seeds (5-fold cross-validation), compared to 83.1% for the best fixed topology (star). Correctness labels for training use majority vote outcomes rather than Stackelberg-weighted decisions, avoiding contamination from optimized θ^* parameters.

6 Theoretical Analysis

Theorem 1 (Existence of Stackelberg Equilibrium). *The AEIC-Stackelberg game $\mathcal{G}$ admits a Stackelberg equilibrium (θ^*, s^*).*

Proof. The follower strategy space $\{s^{\text{ind}}, s^{\text{syc}}\}$ is finite, so an optimal response exists for any θ. The parameter space Θ is compact (bounded). The leader's objective $\text{Accuracy}(\theta) + \lambda \cdot \text{IC}(\theta)$ is bounded on Θ. By the Weierstrass extreme value theorem, the maximum is attained at some $\theta^* \in \Theta$. Together with the followers' best responses, this constitutes a Stackelberg equilibrium.

Theorem 2 (ϵ-Incentive Compatibility). *For any $\epsilon \in (0, 1)$, setting $\lambda \geq \frac{1}{1-\epsilon}$ in Eq. (7) guarantees IC$(\theta^*) \geq \epsilon$.*

Proof. We first establish feasibility: there exists $\theta_0 \in \Theta$ with $\mathrm{IC}(\theta_0) = 1$ (set IC/EQ exponents large, peer pressure $\gamma = 0$, making weights strictly monotone in quality).

By optimality of θ^*:

$$\mathrm{Acc}(\theta^*) + \lambda \cdot \mathrm{IC}(\theta^*) \geq \mathrm{Acc}(\theta_0) + \lambda \cdot 1 \tag{16}$$

Rearranging with $\mathrm{Acc}(\theta_0) - \mathrm{Acc}(\theta^*) \geq -1$:

$$\mathrm{IC}(\theta^*) \geq 1 - \frac{1}{\lambda} \tag{17}$$

Setting $\lambda \geq \frac{1}{1-\epsilon}$ yields $\mathrm{IC}(\theta^*) \geq \epsilon$.

Theorem 3 (Individual Rationality and Welfare). *(a) Under θ^*, all agents achieve non-negative utility when $R \geq C_{ind}/\min_i w_i$.*
(b) Accuracy$(\theta^*) \geq$ Accuracy$(\theta_{uniform})$ for sufficiently small λ.

Proof. (a) Since $w_i \in [0, 1]$ and $C(s_i) \leq C_{\mathrm{ind}}$, choosing $R \geq C_{\mathrm{ind}}/\min_i w_i$ ensures $U_i \geq 0$.

(b) Uniform voting corresponds to $\theta_{\mathrm{unif}} \in \Theta$. By optimality: $\mathrm{Acc}(\theta^*) + \lambda \cdot \mathrm{IC}(\theta^*) \geq \mathrm{Acc}(\theta_{\mathrm{unif}}) + \lambda \cdot \mathrm{IC}(\theta_{\mathrm{unif}})$. For $\lambda \to 0$, this directly yields $\mathrm{Acc}(\theta^*) \geq \mathrm{Acc}(\theta_{\mathrm{unif}})$.

Convergence of Distributed Consensus. Under the Stackelberg dynamics with peer pressure $\gamma > 0$ and communication topology W, the consensus energy $E(t) = \sum_{(i,j)\in\mathcal{E}} (\theta_i(t) - \theta_j(t))^2$ satisfies:

$$E(t) \leq E(0) \cdot \exp\left(-\frac{2\gamma t}{n-1}\right) \tag{18}$$

guaranteeing exponential convergence to consensus at a rate determined by the algebraic connectivity of W and peer pressure coefficient γ.

7 Experiments

7.1 Experimental Setup

Datasets. We evaluate on three diverse reasoning benchmarks: MMLU-Hard (graduate-level knowledge), StrategyQA (commonsense with implicit reasoning), and TruthfulQA (factual accuracy under adversarial framing). Each dataset contributes 300 questions, yielding 900 questions total.

Agent Configuration. We deploy $n = 5$ DeepSeek-V3 [7] instances under each of five communication topologies (parallel, sequential, star, debate, hierarchical), generating 22,500 AEIC-structured responses (5 agents $\times$ 900 questions $\times$ 5 topologies). Since question sets were sampled independently per topology, 633 of 900 questions overlap across all five topologies (161/300/172 for MMLU-Hard/StrategyQA/TruthfulQA). Cross-topology comparisons (router, oracle) use this intersection; per-topology baselines use the full 300 questions.

Signal Computation. Quality signals are computed using sentence embeddings (paraphrase-multilingual-MiniLM-L12-v2). The topology-aware adjacency matrix W determines neighbor sets for EV and agreement computation.

Optimization. Per-topology θ_T^* is optimized via differential evolution (16 parameters, population 16, 30 generations) and evaluated using 5-fold stratified cross-validation to report honest held-out accuracy (preventing train/test leakage). Router evaluation also uses 5-fold CV with five random seeds for stability estimation.

Baselines.

- **Single Agent**: Agent 0's conclusion without aggregation.
- **ChatEval** [3]: Majority voting on parallel topology (our reimplementation).
- **Multi-round Debate** [8]: Majority voting after 3 rounds of debate (our reimplementation).
- **AgentVerse** [5]: Solver–reviewer pattern with central decision (our reimplementation, 200 questions per dataset).
- **MajVote**: Simple majority voting on each topology (5 variants).
- **Weighted Vote**: IC-weighted and agreement-weighted voting without Stackelberg.
- **AEIC+SG**: Stackelberg-weighted voting with per-topology θ^* (5 variants).

All baselines use the same LLM backend (DeepSeek-V3) and the same cached responses, ensuring fair comparison. † = our reimplementation.

7.2 Main Results

Table 1 presents the complete comparison. Figure 1 provides a visual summary.

Key observations: (1) MajVote (star) is the best fixed-topology method at 82.7%, outperforming all AEIC+SG variants even after cross-validation (best AEIC+SG is star at 81.7%). (2) The adaptive router achieves 87.6%, +4.5pp over the best fixed topology on aligned questions. (3) The oracle ceiling is 92.9%, with a remaining 5.3pp gap. (4) Per-dataset analysis reveals that AEIC+SG underperforms MajVote most on commonsense reasoning: on StrategyQA, AEIC+SG star achieves 79.3% vs. MajVote star's 80.3%, while the gap is smallest on knowledge-intensive MMLU-Hard (84.3% vs. 85.7%). (5) AEIC+SG CV test accuracy closely matches MajVote across topologies (within 1pp on 4 of 5), confirming that quality-weighted voting offers limited advantage over simple majority voting when agents share the same LLM backend.

7.3 AEIC-Weighted Voting vs. Majority Voting

Figure 2 reveals a surprising finding: majority voting outperforms AEIC+SG weighted voting on 4 of 5 topologies.

With honest cross-validated evaluation, AEIC+SG performs comparably to MajVote: parallel 77.8% vs. 77.6%, sequential 75.3% vs. 74.2%, star 81.7% vs.

Table 1. Consensus accuracy (%) across three datasets. Per-topology baselines use 300 questions each; router and oracle are evaluated on 633 questions common to all topologies (161/300/172 for MMLU-Hard/StrategyQA/TruthfulQA). Best fixed-topology in *italic*, overall best in **bold**.

Method	MMLU-Hard	StrategyQA	TruthfulQA	Average
Single-agent and voting baselines				
Single Agent (N=1)	65.3	79.0	80.0	74.8
MajVote (parallel)†	69.7	81.7	81.3	77.6
MajVote (sequential)	79.0	71.0	72.7	74.2
MajVote (star)	*85.7*	*80.3*	*82.0*	*82.7*
MajVote (debate)†	83.7	72.3	75.3	77.1
MajVote (hierarchical)	82.0	74.7	75.0	77.2
AgentVerse†	74.0	63.5	70.0	69.2
Weighted Vote (IC)	70.0	80.3	81.0	77.1
AEIC-Stackelberg (per-topology θ_T^*)				
AEIC+SG (parallel)	71.7	78.0	78.7	$77.8_{\pm 5.6}$‡
AEIC+SG (sequential)	74.7	69.0	70.0	$75.3_{\pm 2.5}$‡
AEIC+SG (star)	84.3	79.3	82.0	$81.7_{\pm 3.4}$‡
AEIC+SG (debate)	80.0	72.3	72.0	$76.3_{\pm 3.0}$‡
AEIC+SG (hierarchical)	84.7	76.3	77.0	$77.3_{\pm 3.5}$‡
Adaptive routing (ours, 5-fold CV)				
Neural Router (shared)	91.3	86.0	84.3	$86.9_{\pm 0.5}$
Neural Router (per-dataset)	**92.5**	**86.2**	**85.6**	**87.6**$_{\pm 0.3}$
Oracle (any of 5 topologies)	97.5	91.0	91.9	92.9

† Our reimplementation using DeepSeek-V3. ‡ Average column reports 5-fold CV test accuracy ± std; per-dataset columns use full-data θ^*.

82.7%, debate 76.3% vs. 77.1%, hierarchical 77.3% vs. 77.2%. McNemar tests on the full 900 questions (per-topology θ^* trained on all data) confirm that AEIC+SG is not significantly better than MajVote on any topology except hierarchical (p=0.017), and is in fact significantly worse on parallel (p=0.010), sequential (p=0.004), and debate (p=0.003). This confirms that AEIC signals measure *reasoning process quality, not answer correctness*: a coherent wrong answer scores as highly as a correct one.

This motivates using AEIC signals for *meta-level topology selection* rather than direct vote weighting—they capture topology-question interactions that are invisible to single-scalar quality scores.

7.4 Learned Parameter Analysis

Figure 3 shows the optimized θ^* parameters across five topologies.

Three patterns emerge: (1) **Anti-sycophancy**: Star and debate learn $w_{\text{EV}} \approx 0.01$, discarding cross-agent consensus where agents observe others' outputs, making EV a proxy for conformity. (2) **High consensus pressure on chains**:

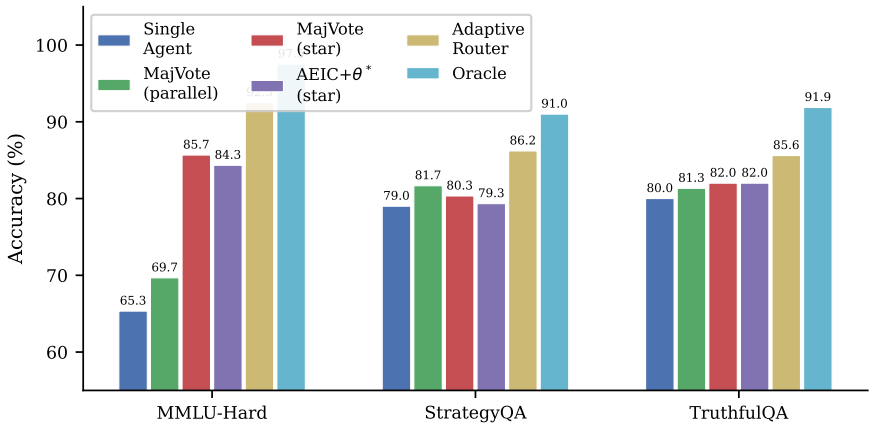


Fig. 1. Accuracy comparison across three datasets. The adaptive router outperforms all fixed-topology methods.

Sequential ($\lambda = 4.80$) and hierarchical ($\lambda = 4.69$) enforce strong agreement to counteract directional information flow. (3) **Elevated weights on hierarchical**: All signal weights are substantially higher (0.45–0.90 vs. 0.01–0.36 elsewhere), reflecting multi-level aggregation where every quality dimension helps identify reliable intermediaries.

7.5 Router Performance Progression

Figure 4 shows the accuracy progression from single-agent to oracle.

The shared router (86.9%) improves substantially over the best fixed topology on aligned questions (star, 83.1%). Per-dataset specialization adds +0.7pp (87.6%), closing 46.5% of the gap to the oracle ceiling (92.9%). All router numbers use top-1 selection with 5-fold cross-validation.

Per-dataset analysis. The router's advantage varies across datasets: on MMLU-Hard, the per-dataset router achieves 92.5% vs. star's 90.1% (+2.4pp); on StrategyQA, 86.2% vs. parallel's 81.7% (+4.5pp); on TruthfulQA, 85.6% vs. star's 81.4% (+4.2pp). The largest improvement on StrategyQA reflects that commonsense reasoning benefits most from adaptive topology selection, as the gap between best and worst fixed topology is largest (81.7% parallel vs. 71.0% sequential).

Why routing helps. Figure 7 shows the per-topology accuracy heatmap on aligned questions. No single topology dominates: star leads on MMLU-Hard (90.1%) and TruthfulQA (81.4%), but parallel is best for StrategyQA (81.7%). The oracle ceiling (∼93%) requires selecting different topologies for different questions—exactly what the router learns to do.

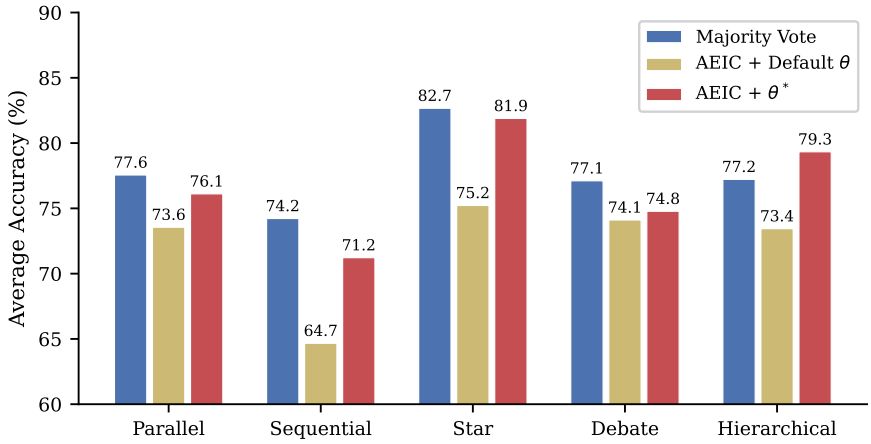


Fig. 2. Per-topology comparison averaged over three datasets. AEIC+θ^* only outperforms majority voting on hierarchical topology.

Topology agreement. On the 633 aligned questions, 55.1% are answered correctly by all five topologies and 7.0% by none—routing has no impact on these 62.1%. The router's value comes from the remaining 37.9% where topologies disagree, particularly the 9.0% of "hard" questions where only 1–2 topologies succeed (Figure 5).

Routing error analysis. The shared router selects each topology with roughly balanced frequency (16–29%), with star most frequently selected (28.6%). Of 83 router errors (13.1%), 44 occur on questions where *no* topology produces correct consensus—inevitable misses. Only 39 errors (6.2%) are routing mistakes where a correct topology existed but was not selected.

Router feature importance. Feature group ablation (zeroing each group and measuring accuracy change, Figure 6) reveals that *voting features* (majority ratio, yes/no distribution) are the most important for routing (−2.1pp when ablated), followed by per-agent EV signals (−0.3pp). Interestingly, removing EQ-related features (per-agent EQ and EQ/EV summary statistics) *improves* router accuracy by up to 1.6pp, paralleling the AEIC+SG ablation finding. This suggests that evidence quality is noisy across both direct weighting and meta-level routing.

7.6 Signal Ablation

Figure 8 shows the contribution of each quality signal to AEIC+SG under star topology.

EC is the most critical signal (−1.15pp). IC and EV each contribute ∼0.5pp. Removing EQ *improves* accuracy by 0.40pp—evidence quality assessment introduces noise in some domains.

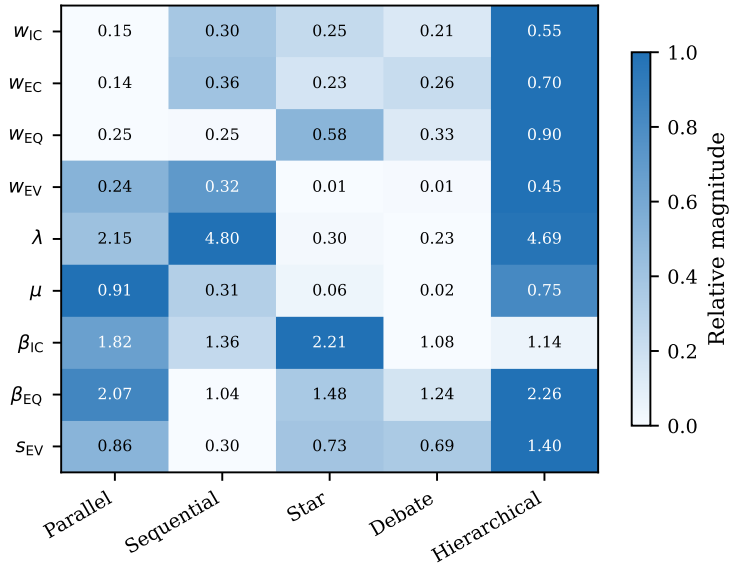


Fig. 3. Optimized θ^* parameters across five topologies (row-normalized). Star and debate learn near-zero w_{EV} (anti-sycophancy). Hierarchical learns high λ and elevated signal weights.

7.7 Online Stackelberg Feedback

We evaluate the online Stackelberg mechanism (Section 4.5) on debate and hierarchical topologies, comparing standard execution against online feedback where quality signals are injected between rounds. Table 2 shows results on 30 questions per dataset.

The results reveal a nuanced picture: online feedback produces a large improvement on MMLU-Hard debate (+6.7pp), where the iterative debate structure allows agents to incorporate quality assessments and refine their reasoning. However, the effect is mixed on other dataset–topology combinations, with no overall improvement on debate (80.0% both) and a decline on hierarchical (−4.4pp). Expanded validation with larger matched samples (n=41–74) confirms the pattern: debate benefits on MMLU-Hard (+5.4pp, n=74) and StrategyQA (+2.4pp, n=41), but degrades on TruthfulQA (−6.7pp); hierarchical never improves (−5.4pp average). The degradation on TruthfulQA suggests that quality feedback may encourage over-confidence on adversarially framed questions. These results highlight that online incentive shaping is most effective when the topology supports genuine iterative refinement (debate) rather than one-directional information flow (hierarchical), and when the task rewards deeper reasoning (MMLU-Hard) rather than resisting misleading cues (TruthfulQA).

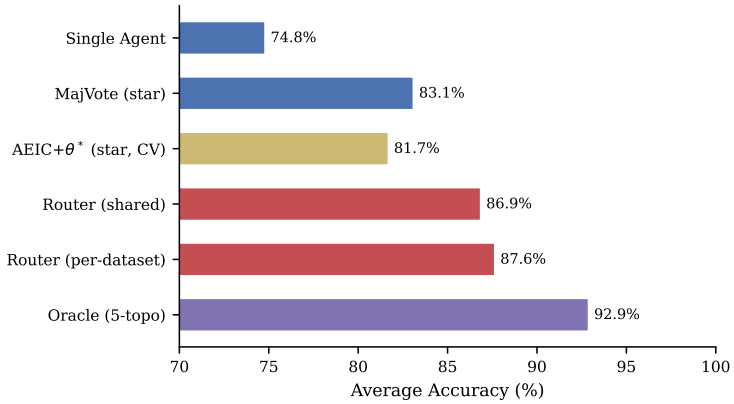


Fig. 4. Accuracy progression from single-agent to oracle. The per-dataset router closes 46.5% of the gap between best fixed topology and oracle.

Table 2. Online vs. offline Stackelberg feedback (MajVote accuracy, %, 30 questions per dataset). Online mode injects quality signals between rounds/phases.

Condition	MMLU-Hard	StrategyQA	TruthfulQA	Average
Debate (offline)	80.0	76.7	83.3	80.0
Debate (online)	**86.7**	76.7	76.7	80.0
Δ (online−offline)	+6.7	0.0	−6.7	0.0
Hierarchical (offline)	76.7	80.0	83.3	80.0
Hierarchical (online)	76.7	73.3	76.7	75.6
Δ (online−offline)	0.0	−6.7	−6.7	−4.4

Offline uses cached responses; online uses live DeepSeek-V3 calls with inter-round quality feedback.

7.8 Accuracy–Incentive Trade-off

The optimized θ^* configurations naturally achieve high incentive compatibility without explicit IC constraints: IC rates of 98.9% (parallel), 94.0% (sequential), 98.4% (star), and 100.0% (debate). Hierarchical is the exception at 24.2%, reflecting its elevated signal weights that prioritize multi-level aggregation over strict quality ordering. This suggests that accuracy optimization alone produces near-IC mechanisms—the Stackelberg weight structure inherently rewards higher-quality reasoning. The λ term (Theorem 2) provides a formal guarantee for adversarial settings but is largely redundant when agents share the same LLM backend.

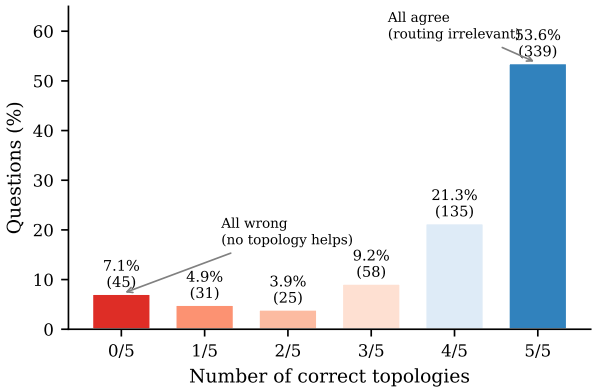


Fig. 5. Distribution of correct topologies per question (633 aligned). The router's impact concentrates on the 37.9% where topologies disagree (1–4 correct).

8 Discussion

Why Stackelberg over Nash? Yi et al. [25] model multi-agent debate as a simultaneous-move Bayesian game. Our Stackelberg formulation is more natural for mechanism design: the aggregation rule is *committed to* before agents act, enabling the leader to shape follower behavior through credible commitment—a property absent in Nash equilibrium where all players move simultaneously.

Quality Signals vs. Correctness. Our most significant finding is that AEIC quality signals measure reasoning process quality, not answer correctness. McNemar tests confirm that AEIC+SG weighted voting is significantly *worse* than majority voting on three topologies (parallel p=0.010, sequential p=0.004, debate p=0.003) and only significantly better on hierarchical (p=0.017). A coherent, well-evidenced wrong answer scores as highly as a correct one. This implies that quality-based weighting cannot substitute for diversity-based aggregation when agents share the same LLM backend and similar knowledge biases.

AEIC as Feature Extraction. Despite limited value as direct voting weights, AEIC signals are informative features for topology routing. Feature ablation reveals that voting features (majority ratio, yes/no distribution) contribute the most to routing accuracy (−2.1pp), while EQ-related features are harmful (removing them *improves* accuracy by up to 1.6pp). This parallels the AEIC+SG ablation findings: evidence quality signals introduce noise at both the weighting and routing levels. The router achieves 87.6% using these features with top-1 selection, suggesting that the appropriate level for quality assessment in multi-agent systems is *meta-level topology selection* rather than agent-level vote weighting.

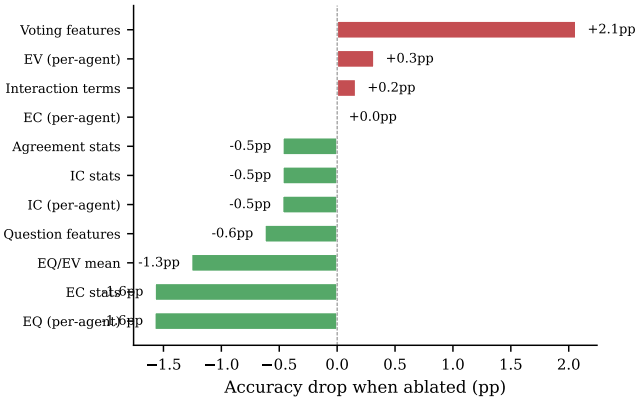


Fig. 6. Router feature group importance via leave-one-out ablation. Positive drop means accuracy decreases when ablated (feature is useful). Voting features are most important; EQ features are harmful.

Anti-Sycophancy from First Principles. Optimal mechanisms learn near-zero EV weights on star and debate topologies, providing a principled anti-sycophancy mechanism. On information-sharing topologies, high EV reflects social influence rather than independent verification. This emerges from accuracy optimization without explicit constraints, complementing heuristic approaches [16] with a game-theoretic foundation.

Oracle Gap Analysis. The 5.3pp gap between routing (87.6%) and oracle (92.9%) arises from questions where no topology produces correct consensus (7.0%) and routing errors where the selected topology fails despite a correct alternative existing. Analyzing topology agreement: on 62.1% of questions, all five topologies agree (55.1% all correct, 7.0% all wrong); routing adds no value here. The router's impact comes from the 9.0% of "hard" questions where only 1–2 topologies succeed. Our router closes 46.5% of the gap between the best fixed topology (83.1%) and the oracle. The remaining gap could be reduced by richer topology spaces, heterogeneous agent pools, or improved routing features that better distinguish question difficulty patterns.

Comparison with RL-Based Adaptive Topology. RUMAD [1] uses reinforcement learning to adaptively control debate rounds and agent participation within a single topology type. CARD [15] routes across debate configurations. Our approach differs in three ways: (1) we select among five fundamentally different communication structures (parallel, sequential, star, debate, hierarchical), not just debate variants; (2) our routing is based on interpretable AEIC quality signals with game-theoretic grounding, while RUMAD uses opaque RL policies; (3) our Stackelberg mechanism provides formal incentive compatibility guarantees (Theorem 2). The approaches are complementary—within-topology adaptation (RUMAD) and cross-topology selection (ours) could be combined.

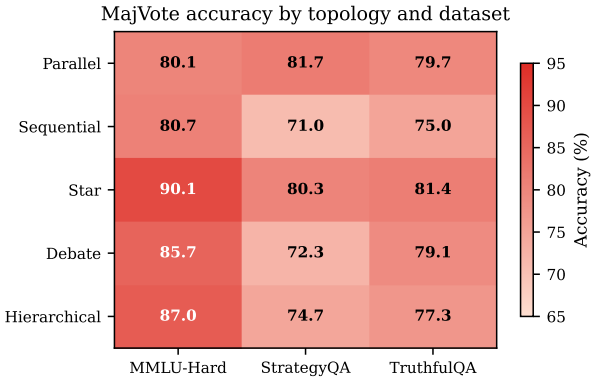


Fig. 7. MajVote accuracy by topology and dataset on aligned questions. No single topology dominates, motivating adaptive routing.

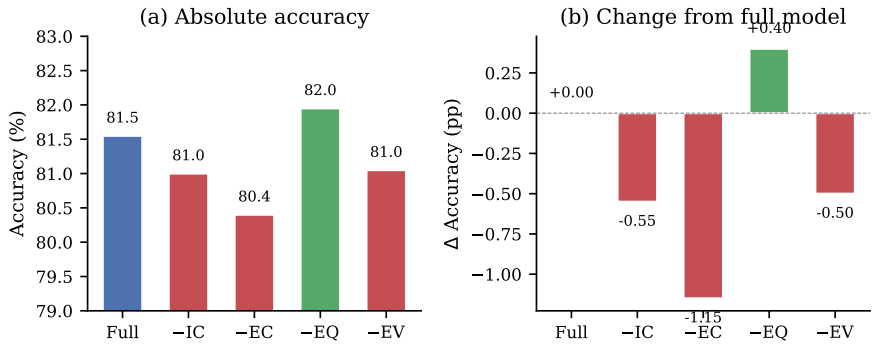


Fig. 8. Signal ablation under star topology. EC is the most critical signal (−1.15pp). Removing EQ slightly improves accuracy, consistent with its low optimized weight.

Online Stackelberg Game. Our online extension injects quality feedback between rounds, transforming Stackelberg from post-hoc reweighting to real-time incentive shaping. The experimental results (Table 2) reveal that effectiveness depends critically on topology–task interaction: debate benefits strongly on knowledge-intensive tasks (+6.7pp on MMLU-Hard, confirmed at +5.4pp with n=74) and shows moderate gains on commonsense reasoning (+2.4pp on StrategyQA, n=41); hierarchical shows no improvement (−5.4pp average with expanded validation). This suggests that online feedback is most valuable when (1) the topology supports genuine iterative refinement and (2) the task rewards deeper reasoning rather than resistance to misleading cues. The mixed results motivate adaptive feedback policies that modulate signal injection based on task characteristics.

Limitations. (1) The router requires running all five topologies for feature extraction (5× cost); online deployment needs topology prediction from question

features alone. (2) All agents use the same LLM (DeepSeek-V3); heterogeneous pools may behave differently. (3) Online Stackelberg feedback shows mixed results: beneficial on MMLU-Hard debate but degrading on TruthfulQA and hierarchical, suggesting the need for task-adaptive feedback policies. (4) The online evaluation uses 30–74 questions per condition (API availability limited some conditions), with the largest effects (±5–7pp) reaching marginal significance. (5) Question sets are not fully shared across topologies (only 633 of 900 questions overlap across all five), limiting the router evaluation set.

9 Conclusion

We presented an adaptive game-theoretic framework for multi-agent LLM consensus that combines AEIC reasoning decomposition, Stackelberg incentive mechanisms with online feedback, and neural topology routing. On 633 questions evaluated across all five communication topologies on three reasoning benchmarks with 5 agents, our per-dataset router achieves 87.6% accuracy (+4.5pp over the best fixed topology on aligned questions), demonstrating that AEIC-derived features enable effective adaptive topology selection.

Our key findings include: (1) AEIC quality signals measure reasoning process quality rather than answer correctness—AEIC+SG is significantly worse than majority voting on 3 of 5 topologies (McNemar $p<0.05$)—making them more valuable as topology routing features than as direct voting weights; (2) per-topology Stackelberg optimization discovers interpretable parameter profiles, including anti-sycophancy effects where cross-agent consensus signals are suppressed on information-sharing topologies; (3) the mechanism naturally achieves high incentive compatibility (94–100% on 4 of 5 topologies) without explicit IC constraints; and (4) online Stackelberg feedback between rounds can substantially improve debate accuracy on knowledge-intensive tasks (+5.4pp on MMLU-Hard with n=74), though the effect is topology- and task-dependent.

Future work includes developing online routing that selects topologies from question features without running all candidates, evaluating the online Stackelberg game with live LLM agents, exploring heterogeneous agent pools with diverse LLM backends, and combining cross-topology selection with within-topology adaptation [1].

Disclosure of Interests. The authors have no competing interests to declare.

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